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Study on the Effect of Bacillus subtilis Bacteria on the Mechanical, Durability and Self-Healing Properties of Concrete

Hanumant Sharan Singh
Maharishi Mahesh Yogi Ramayan University, Ayodhya (UP)
hanumantsharansingh@gmail.com

Abstract

Concrete is essential, and it is used in large quantities in almost all building methods. However, cracking is an inherent problem in the concrete and can pass and absorb water, chlorides and other aggressive substances to penetrate the concrete matrix, reduce durability, and require large maintenance. Bacterial concrete is an emerging self-healing technology in which microorganisms precipitate calcium carbonate within cracks. *Bacillus subtilis* is particularly interesting because it can form resistant endospores and can survive the highly alkaline environment associated with cementitious materials. This paper reviews the effect of *B. subtilis* on concrete and mortar with particular emphasis on compressive strength, tensile strength, flexural strength, water absorption and crack-healing capacity. The mechanism is primarily associated with microbial-induced calcium carbonate precipitation (MICP), which deposits mineral phases within cracks and pores. Although the results are promising, bacterial concentration, nutrient availability, encapsulation, curing conditions, crack width and long-term bacterial viability require further investigation before widespread field application.

Keywords: Bacillus subtilis, bacterial concrete, bio-concrete, self-healing concrete, MICP, calcium carbonate, crack healing, concrete durability.

1. Introduction

Concrete is an essential material in civil engineering and is extensively used in buildings, bridges, pavements, dams, tunnels and other infrastructure. Despite its excellent compressive strength, concrete is relatively weak in tension and is susceptible to cracking. Cracks may develop because of drying shrinkage, thermal effects, loading, settlement and environmental exposure. Once interconnected cracks develop, water and aggressive ions can enter the concrete and accelerate deterioration.

Traditional crack-repair techniques include cement grouting, epoxy injection, polymer treatment and surface sealants. These methods generally require identification of cracks and human intervention. Consequently, researchers have investigated **self-healing concrete**, in which the material can seal or repair cracks with limited external intervention.

One promising approach is the incorporation of microorganisms into concrete. Certain bacteria can induce precipitation of calcium carbonate ($CaCO_3$), which fills cracks and pores. This process is generally referred to as microbially induced calcium carbonate precipitation (MICP). Reviews of bacterial concrete identify *Bacillus* species as important because their spores can tolerate the highly alkaline and physically restrictive concrete environment.

2. Research Objectives

The main objectives of this research paper are:

1. To study the suitability of *Bacillus subtilis* for bacterial concrete.
2. To understand the mechanism of microbial-induced calcium carbonate precipitation.
3. To investigate the influence of *B. subtilis* on compressive strength.

4. To study its effect on splitting tensile and flexural strength.
5. To evaluate its influence on water absorption and concrete durability.
6. To study the influence of crack width and curing conditions on healing.

3. Literature Review

The Indian government has recently proposed a large programme to expand the country's network of motorways, power plants, and other essential facilities. Concrete has assumed its appropriate place in the construction of buildings and other structures, and a vast quantity of concrete is being used to fulfil the demands of globalization [1-3].

3.1 Bacterial Self-Healing Concrete

Bacterial self-healing concrete incorporates microorganisms capable of producing mineral precipitates inside cracks. The precipitated calcium carbonate can act as a filling material and reduce crack width and permeability.

3.2 Selection of *Bacillus subtilis*

Bacillus subtilis is a Gram-positive, spore-forming bacterium. Spore formation is important for concrete applications because the cement matrix presents severe conditions for microorganisms, particularly high alkalinity.

4. Mechanism of Bacterial Crack Healing

The principal mechanism considered in many bacterial-concrete studies is **microbial-induced calcium carbonate precipitation**.

When ureolytic bacteria are supplied with urea, the bacterial urease enzyme promotes urea hydrolysis:

$$CO(NH_2)_2 + 2H_2O \rightarrow 2NH_4^+ + CO_3^{2-}$$

The carbonate ions subsequently react with calcium ions:

$$Ca^{2+} + CO_3^{2-} \rightarrow CaCO_3$$

Simplified mechanism

Bacterial spores/cells → metabolic activity → carbonate production → $CaCO_3$ precipitation → crack filling → reduced permeability

The precipitated material can include calcite and other mineral phases depending on the mixture and curing environment. Microscopic investigation can therefore be used to verify the healing mechanism.

5. Incorporation of *Bacillus subtilis* into Concrete

There are several possible methods for introducing bacteria into concrete.

5.1 Direct incorporation

Bacterial cells or spores are added directly to the mixing water or concrete mixture.

Advantages:

- Simple mixing procedure.
- Relatively low initial processing requirement.

Limitations:

- Bacteria may be exposed directly to the highly alkaline cement environment.
- Survival can decrease with time.

5.2 Encapsulation

Bacterial spores are enclosed within a protective carrier before being incorporated into concrete.

Potential carriers include: Sodium alginate, Silica gel, Lightweight aggregates, Hydrogel systems, Microcapsules

5.3 Immobilization

Bacteria can also be immobilized onto a suitable carrier material. This approach can protect the microorganisms and provide a localized environment for mineral precipitation.

6. Materials

The following materials may be used:

- Ordinary Portland Cement (OPC)
- Fine aggregate
- Coarse aggregate
- Potable water
- *Bacillus subtilis*
- Suitable bacterial nutrient medium
- Calcium source, where required
- Urea, where the selected MICP mechanism requires it
- Encapsulation material such as sodium alginate

7. Tests to be Conducted

7.1 Compressive Strength

Concrete cubes can be tested at:

- 7 days
- 14 days
- 28 days
- 56 days
- 90 days

Compressive strength is calculated as:

$$f_c = \frac{P}{A}$$

Where:

- f_c= compressive strength
- P= failure load
- A= loaded area

Cylindrical specimens can be tested to determine splitting tensile strength.

$$f_t = \frac{2P}{\pi LD}$$

Where:

- P= applied load
- L= cylinder length
- D= cylinder diameter

7.3 Flexural Strength

Beam specimens can be tested using a standard flexural test arrangement.

The test can determine whether bacterial inclusion affects the tensile behavior of concrete under bending.

7.4 Water Absorption

Water absorption is an important durability indicator because cracks and interconnected pores provide pathways for water and aggressive chemicals.

The percentage absorption can be calculated as:

$$Water\ Absorption(\%) = \frac{W_s - W_d}{W_d} \times 100$$

where:

- W_s= saturated weight
- W_d= dry weight.

The 2023 experimental investigation reported lower water absorption in *B. subtilis* bio-mortar than in normal mortar.

8. Crack-Healing Test

Controlled cracks can be introduced into specimens after an appropriate curing period.

Suggested crack widths:

- 0.3 mm
- 0.5 mm
- 1.0 mm

Crack width can be monitored using:

- Crack-width microscope
- Crack comparator
- Digital image analysis

The healing ratio can be calculated as:

$$Healing\ Ratio(\%) = \frac{W_i - W_t}{W_i} \times 100$$

Where:

- W_i= initial crack width
- W_t= crack width after healing period.

The crack should be photographed periodically to document the development of calcium carbonate precipitation.

Effect on Water Absorption

The formation of calcium carbonate can partially fill pores and microcracks.

In the cited experimental study, average water absorption at 28 days was:

- Normal mortar: **3.5%**
- Bio-mortar: **2.1%**

The researchers attributed the reduction partly to MICP-related calcium carbonate precipitation that filled defects in the mortar matrix.

This is important because lower water absorption can reduce the transport of moisture and potentially aggressive substances through the concrete.

Effect on Crack Healing

The most important advantage of bacterial concrete is its potential to heal cracks autonomously.

Microstructural Investigation

Microstructural analysis is important for confirming bacterial mineralization.

Recommended techniques

Scanning Electron Microscopy (SEM):

- Examines crack-filling products.
- Shows morphology of precipitated minerals.
- Helps identify the relationship between bacteria, capsules and cement matrix.

Energy Dispersive X-ray Spectroscopy (EDS):

- Identifies elements such as Ca, C and O.
- Supports identification of calcium-carbonate-rich deposits.

X-Ray Diffraction (XRD):

- Can identify crystalline mineral phases such as calcite.

Environmental Considerations

Bacterial concrete has potential environmental advantages because improved durability and crack sealing may extend service life and reduce repair requirements. However, the environmental performance depends on the bacterial growth medium, nutrients, carrier materials and production process.

In particular, reviews have noted that the environmental impact of MICP systems can be influenced by the amount and type of urea used.

Therefore, bacterial concrete should not automatically be considered environmentally superior in every formulation; its complete life cycle should be evaluated.

Applications

Potential applications include:

- Building structures
- Bridge decks
- Water-retaining structures
- Underground structures
- Tunnels
- Pavements
- Drainage structures
- Precast concrete products
- Marine and coastal infrastructure
- Concrete repair systems

The greatest potential is likely in applications where small cracks develop and where preventing water ingress can provide significant durability benefits.

9. Results and Discussion Based on previously published research, the proposed investigation is expected to examine whether *B. subtilis*:

- Increases compressive strength.
- Improves splitting tensile strength.
- Improves flexural strength.
- Reduces water absorption.
- Reduces crack width through $CaCO_3$ precipitation.
- Improves the durability-related properties of concrete.
- Produces observable mineral deposits inside cracks.

The available literature indicates that *B. subtilis* has considerable potential as a biological agent for self-healing cementitious materials. Its spore-forming ability is particularly useful because bacterial spores can withstand environmental conditions that may be unfavorable to actively growing cells.

The improvement in mechanical properties reported in some experiments may be associated with pore refinement and mineral precipitation. The reduction in water absorption provides additional evidence that the bacterial treatment can modify the pore/crack structure.

The most significant finding is the ability to produce calcium carbonate within artificially generated cracks. However, bacterial concrete should not be viewed simply as concrete containing bacteria. Its performance is governed by an interconnected system involving the bacterial strain, concentration, nutrient availability, calcium source, carrier or encapsulation method, crack width, moisture and curing conditions.

10. Conclusion

The study of *Bacillus subtilis* in concrete demonstrates the potential of biological processes to improve the performance and durability of cement-based materials. The major mechanism is microbial-induced calcium carbonate precipitation, through which bacterial activity promotes formation of mineral deposits that can fill cracks and pores.

Published experimental research has demonstrated improvements in mechanical properties and reductions in water absorption in *B. subtilis* bio-mortar. The study reported 7–8.9% increases in compressive strength, 7.4–13.33% increases in splitting tensile strength and 4.7–7% increases in flexural strength for its investigated mixtures. It also reported substantial crack-healing ratios after 28 days.

The technology therefore represents a promising research direction for durable and potentially lower-maintenance concrete. Further research should focus on optimum bacterial dosage, encapsulation technology, nutrient systems, long-term viability, repeated healing cycles, durability under realistic exposure conditions, environmental impacts and economic feasibility.

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